# Inya-CloneShield Solution Overview

Real-Time Indic Voice Biometrics & AI Deepfake Defense for Indian Banking Telephony

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## 01. Problem Statement & Threat Vector

Conversational AI banking bots are booming in India, but traditional IVRs authenticate only **knowledge** (Account #, OTP), not **biological identity**.

- **AI Synthetic Voice Clones:** Neural vocoders (HiFi-GAN, ElevenLabs) clone victim voices from just 3-second audio clips.
- **The "Two Humans" Blindspot:** Impostor friends or relatives recite genuine customer details over phone calls, bypassing knowledge tests.

## 02. The Inya-CloneShield Solution

A dual-layer, zero-trust voice biometric engine executing directly over 8kHz narrowband telephone channels in **<265 milliseconds**:

- **Layer 1 (Biological Liveness):** Analyzes pharyngeal vocal tract micro-tremors to eliminate AI synthetic vocoder clones (12.4% liveness).
- **Layer 2 (7-Band Formant Timbre):** Matches 7 distinct resonant spectral formant cavities against enrolled user baselines with >96% accuracy.

## 03. End-to-End Operational Workflow

| 1. Bank Selection | 2. Quick Enroll | 3. Live Mic Call | 4. Instant Audit | 5. Auto Defense |
|---|---|---|---|---|
| HDFC, SBI, ICICI, APGB | 5-Sec Voiceprint Capture | F0 & Formant Extraction | 265ms Biometric Decision | Voice Bot or Account Freeze |

## 04. Biometric Verification & Response Matrix

| Caller Scenario | Acoustic Measurement | Trust Score | Security Action | Voice Bot Response |
|---|---|---|---|---|
| **Genuine Customer (Karthik)** | Resonant Timbre Match: **96.8%** | **98 / 100** | Access Authorized + Balance SMS | "Hello Karthik! Your details have matched. Balance is ₹35,000." |
| **Human Impostor (Friend/Scammer)** | Timbre Formant Distance: **32.5%** | **34 / 100** | Access Blocked + Fraud Alert SMS | "Access Denied! Your voice does not match registered customer." |
| **AI Synthetic Deepfake Clone** | Vocoder Artifacts (Liveness: **12.4%**) | **12 / 100** | Critical Intrusion Blocked | "Security Warning! AI synthetic voice clone spoof attack detected." |
| **Emergency Protocol** | Operator or User Single-Click | **LOCKED** | RBI Card Protection Switch Activated | Immediate Account Freeze + Suspended Outgoing Transactions |

## 05. Speed & Telephony SLA

- **Total Latency:** 265ms (<350ms telecom standard).
- **Narrowband Codec:** 8kHz G.711 Telephony.
- **Noise Resilience:** Filtered up to 10dB Mumbai traffic noise.

## 06. Multi-Channel 2FA

- **Voice-OTP Telephony:** Out-of-band call challenge.
- **Instant SMS Dispatch:** Real-time alerts to registered mobile.
- **1-Click Account Freeze:** Immediate transaction suspension.

## 07. Enterprise Compliance

- **1:N Multi-Speaker:** Multi-user bank audit matrix.
- **Forensic Certificate:** Cryptographic timestamped proof.
- **Regulatory Ready:** Aligned with RBI cyber security guidelines.